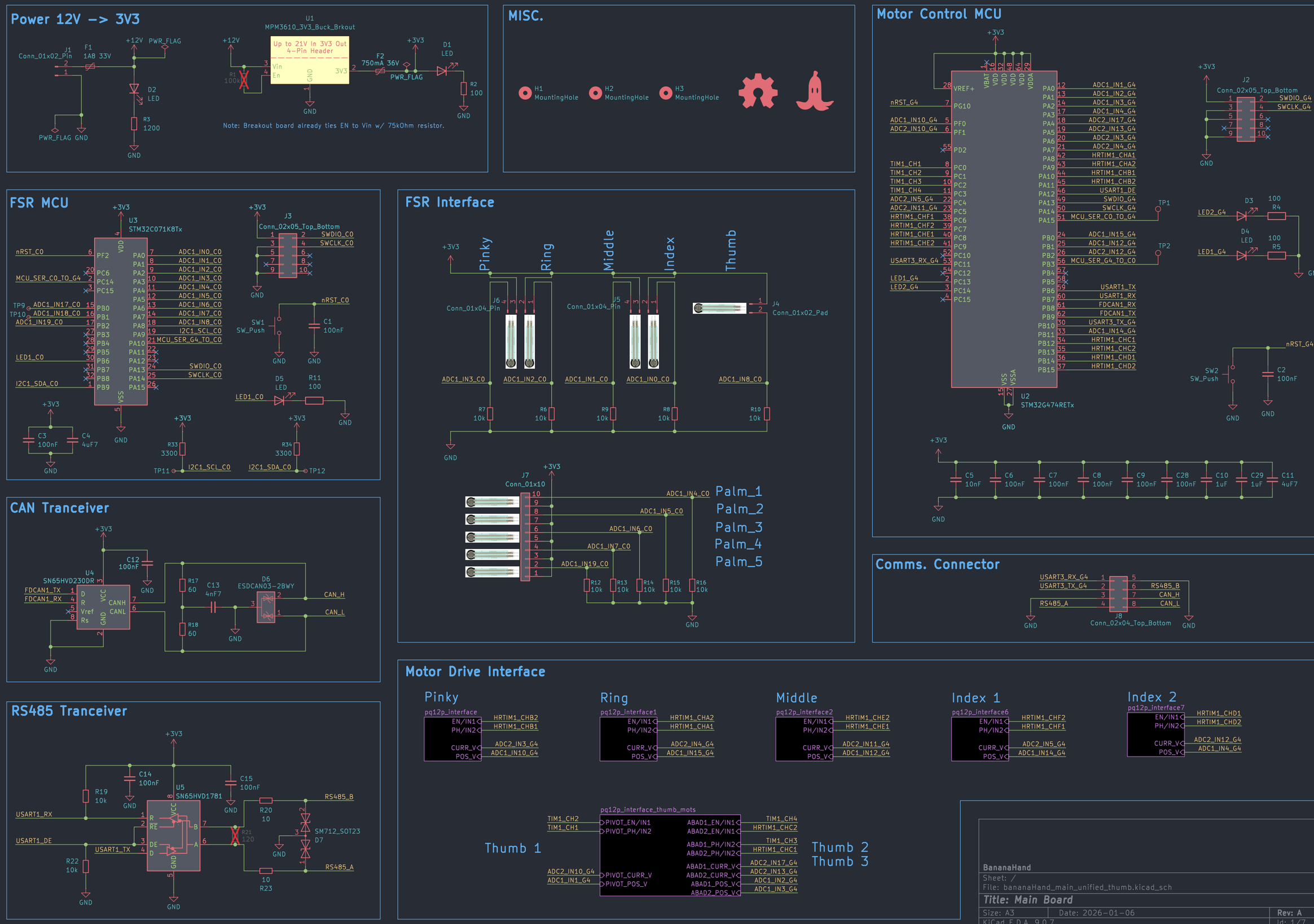
Power 12V -> 3V3
J1
Conn_01x02_Pin
F1
1A8 33V
+12V
PWR_FLAG
D2
LED
R3
1200
PWR_FLAG GND
GND
U1
MPM3610_3V3_Buck_Brkout
Up to 21V In 3V3 Out
4-Pin Header
Vin
En
GND
3V3
R1
100k
F2
750mA 36V
+3V3
D1
LED
R2
100
Note: Breakout board already ties EN to Vin w/ 75kOhm resistor.
MISC.
H1
MountingHole
H2
MountingHole
H3
MountingHole
Motor Control MCU
U2
STM32G474RETx
VREF+
VBAT
VDD
VDDA
VSS
VSSA
nRST_G4
ADC1_IN10_G4
ADC2_IN10_G4
TIM1_CH1
TIM1_CH2
TIM1_CH3
TIM1_CH4
ADC2_IN5_G4
ADC2_IN11_G4
HRTIM1_CHF1
HRTIM1_CHF2
HRTIM1_CHE1
HRTIM1_CHE2
USART3_RX_G4
LED1_G4
LED2_G4
ADC1_IN1_G4
ADC1_IN2_G4
ADC1_IN3_G4
ADC1_IN4_G4
ADC2_IN17_G4
ADC2_IN13_G4
ADC2_IN3_G4
ADC2_IN4_G4
HRTIM1_CHA1
HRTIM1_CHA2
HRTIM1_CHB1
HRTIM1_CHB2
USART1_DE
SWDIO_G4
SWCLK_G4
MCU_SER_C0_TO_G4
TP1
TP2
ADC1_IN15_G4
ADC1_IN12_G4
ADC2_IN12_G4
MCU_SER_G4_TO_C0
USART1_TX
USART1_RX
FDCAN1_RX
FDCAN1_TX
USART3_TX_G4
ADC1_IN14_G4
HRTIM1_CHC1
HRTIM1_CHC2
HRTIM1_CHD1
HRTIM1_CHD2
J2
Conn_02x05_Top_Bottom
SWDIO_G4
SWCLK_G4
D3
100
R4
D4
LED
100
R5
SW2
SW_Push
C2
100nF
C5
10nF
C6
100nF
C7
100nF
C8
100nF
C9
100nF
C28
100nF
C10
1uF
C29
1uF
C11
4uF7
FSR MCU
U3
STM32C071K8Tx
VDD
VSS
PF2
PC6
PC14
PC15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
nRST_C0
MCU_SER_C0_TO_G4
ADC1_IN17_C0
ADC1_IN18_C0
ADC1_IN19_C0
LED1_C0
I2C1_SDA_C0
ADC1_IN0_C0
ADC1_IN1_C0
ADC1_IN2_C0
ADC1_IN3_C0
ADC1_IN4_C0
ADC1_IN5_C0
ADC1_IN6_C0
ADC1_IN7_C0
ADC1_IN8_C0
I2C1_SCL_C0
MCU_SER_G4_TO_C0
SWDIO_C0
SWCLK_C0
J3
Conn_02x05_Top_Bottom
SW1
SW_Push
C1
100nF
D5
LED
R11
100
C3
100nF
C4
4uF7
R33
3300
R34
3300
TP11
TP12
FSR Interface
Pinky
Ring
Middle
Index
Thumb
J6
Conn_01x04_Pin
J5
Conn_01x04_Pin
J4
Conn_01x02_Pad
ADC1_IN3_C0
ADC1_IN2_C0
ADC1_IN1_C0
ADC1_IN0_C0
ADC1_IN8_C0
R7
10k
R6
10k
R9
10k
R8
10k
R10
10k
J7
Conn_01x10
ADC1_IN4_C0
ADC1_IN5_C0
ADC1_IN6_C0
ADC1_IN7_C0
ADC1_IN19_C0
Palm_1
Palm_2
Palm_3
Palm_4
Palm_5
R12
10k
R13
10k
R14
10k
R15
10k
R16
10k
CAN Tranceiver
U4
SN65HVD230DR
FDCAN1_TX
FDCAN1_RX
D
R
Vref
Rs
VCC
CANH
CANL
GND
C12
100nF
R17
60
R18
60
C13
4nF7
D6
ESDCAN03-2BWY
CAN_H
CAN_L
Comms. Connector
USART3_RX_G4
USART3_TX_G4
RS485_A
RS485_B
CAN_H
CAN_L
J8
Conn_02x04_Top_Bottom
RS485 Tranceiver
U5
SN65HVD1781
R
RE
DE
D
VCC
GND
A
B
USART1_RX
USART1_DE
USART1_TX
R19
10k
R22
10k
C14
100nF
C15
100nF
R20
10
R21
120
R23
10
RS485_B
RS485_A
SM712_SOT23
D7
Motor Drive Interface
Pinky
pq12p_interface
EN/IN1
PH/IN2
CURR_V
POS_V
HRTIM1_CHB2
HRTIM1_CHB1
ADC2_IN3_G4
ADC1_IN10_G4
Ring
pq12p_interface1
HRTIM1_CHA2
HRTIM1_CHA1
ADC2_IN4_G4
ADC1_IN15_G4
Middle
pq12p_interface2
HRTIM1_CHE2
HRTIM1_CHE1
ADC2_IN11_G4
ADC1_IN12_G4
Index 1
pq12p_interface6
HRTIM1_CHF2
HRTIM1_CHF1
ADC2_IN5_G4
ADC1_IN14_G4
Index 2
pq12p_interface7
HRTIM1_CHD1
HRTIM1_CHD2
ADC2_IN12_G4
ADC1_IN4_G4
Thumb 1
pq12p_interface_thumb_mots
TIM1_CH2
TIM1_CH1
PIVOT_EN/IN1
PIVOT_PH/IN2
ADC2_IN10_G4
ADC1_IN1_G4
PIVOT_CURR_V
PIVOT_POS_V
ABAD1_EN/IN1
ABAD2_EN/IN1
ABAD1_PH/IN2
ABAD2_PH/IN2
ABAD1_CURR_V
ABAD2_CURR_V
ABAD1_POS_V
ABAD2_POS_V
TIM1_CH4
HRTIM1_CHC2
TIM1_CH3
HRTIM1_CHC1
ADC2_IN17_G4
ADC2_IN13_G4
ADC1_IN2_G4
ADC1_IN3_G4
Thumb 2
Thumb 3
BananaHand
Sheet: /
File: bananaHand_main_unified_thumb.kicad_sch
Title: Main Board
Size: A3
Date: 2026-01-06
Rev: A
KiCad E.D.A. 9.0.7
Id: 1/7